
When Evaluation Cues Become the Evaluation: Auditing Purpose-by-Format Probes in Language Models

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Abstract

1 Evaluation protocols can produce highly reproducible internal signals without
2 measuring the construct that their interpretation invokes. We study this failure
3 mode in a frozen purpose-by-format evaluation design. Each payload is crossed
4 between an evaluation and deployment purpose and between benchmark and casual
5 response formats, allowing purpose effects, format offsets, and purpose–format
6 interactions to be estimated separately. On 11,520 rendered items from 320 payload
7 blocks, held-out cue-vocabulary probes decode stated response purpose strongly
8 across Llama-3.1-8B, Llama-3.1-70B, and Llama-3.3-70B. However, format is also
9 highly decodable, threshold accuracy can remain near chance when AUC is high,
10 and a direct 8B Jacobian-lens diagnostic yields much weaker purpose readouts than
11 the conventional probe. An equal-training-size control at 8B preserves purpose
12 decodability, but does not establish a behavioral or causal endpoint. The resulting
13 evidence supports a methodological conclusion: decodability of evaluation-context
14 cues is not evidence of evaluation awareness. We provide the frozen data contract,
15 model revisions, activation hashes, control analyses, and explicit claim boundaries
16 needed to audit this class of evaluation protocol.

17 1 Introduction

18 Benchmarks frequently attach high-level interpretations to low-level measurements. A model that
19 responds differently when told that an answer will be judged may be described as exhibiting evaluation
20 awareness, strategic behavior, or situational reasoning. Each description requires more than a classifier
21 that separates two prompt conditions. It requires a validated construct, a protocol that isolates the
22 construct from formatting and sampling artifacts, and an endpoint connecting the internal signal to
23 behavior or a decision.

24 This paper audits a purpose-by-format evaluation protocol rather than treating its strongest probe
25 score as a discovery. The protocol crosses two factors: stated response purpose (evaluation versus
26 deployment) and response format (benchmark versus casual). The crossing makes it possible to
27 distinguish a format-shared purpose effect from a format offset and an interaction. It also creates
28 direct failure tests: if a purpose signal disappears under format transfer, or if fixed-threshold accuracy
29 disagrees with AUC, the protocol does not support a simple interpretation.

30 Our main result is deliberately narrower than an awareness claim. On a frozen 320-block dataset,
31 purpose is strongly decodable using held-out cue-vocabulary splits in three Llama models. Yet
32 the same representation carries strong format/template information, and a direct 8B Jacobian-lens

33 diagnostic is substantially weaker for purpose than the conventional probe. The appropriate conclusion
 34 is that the protocol measures a reproducible stated-purpose signal, while the intended latent construct
 35 remains underdetermined.

36 Our contributions are: (1) a hash-anchored full320-v3 evaluation freeze with a crossed purpose–format
 37 design; (2) matched observational probes that report AUC and balanced accuracy separately; (3) cue-
 38 excision, transfer, and equal-training-size controls; (4) a model-specific 8B direct-lens diagnostic; and
 39 (5) a claim-audit framework that distinguishes decodability, geometry, internal mediation, behavioral
 40 causality, and confirmation.

41 **2 Target construct and protocol**

42 **2.1 What is measured**

43 We use *stated response purpose decodability* as the primary estimand. A rendered prompt states that
 44 a response will be evaluated or deployed, and the model’s last-prompt-token residual representation
 45 is probed for that label. We do not equate this estimand with evaluation awareness. In particular,
 46 a probe may recover lexical, stylistic, calibration, or task-format information correlated with the
 47 purpose statement.

48 The frozen full320-v3 data contain 11,520 rendered items from 320 payload blocks. Each block is
 49 crossed over purpose $p \in \{E, D\}$ and format $f \in \{B, C\}$, where E/D denote evaluation/deployment
 50 and B/C denote benchmark/casual format. The intended effect-coded decomposition is

$$h_{p,f} = \mu + pA + fB + pfC + \epsilon, \quad (1)$$

51 where A is the format-shared purpose effect, B is the format offset,
 52 and C is the purpose–format interaction. The primary dataset hash is
 53 5528819d74e696d1a595062047a829836dfde1d2a45287b06a5ac1ba5d1b6c67.

54 **2.2 Activation contract**

55 For each model, activations are the zero-based transformer-block output before final model normal-
 56 ization at the last prompt token after applying the user-only chat template with generation prompt.
 57 Inference uses BF16 and saved activations are FP32. The pinned model revisions are Llama-3.1-8B-
 58 Instruct (0e9e39f2), Llama-3.1-70B-Instruct (1605565b), and Llama-3.3-70B-Instruct (6f6073b4).
 59 The complete per-file provenance is recorded in the accompanying HF snapshot and evidence mani-
 60 fest.

61 **2.3 Held-out evaluation**

62 The conventional probe is StandardScaler followed by logistic regression with $C = 0.1$, maximum
 63 3000 iterations, and fixed random state 0. We use five held-out cue-vocabulary folds. The principal
 64 regime trains on all four purpose–format cells and tests within the held-out vocabulary fold. We
 65 also report benchmark-to-casual and casual-to-benchmark transfer. AUC and balanced accuracy are
 66 separate endpoints: AUC is threshold-free, while balanced accuracy uses the fitted decision threshold
 67 and can expose calibration failure.

68 **3 Full320 observational results**

69 Table 1 reports each model’s peak decorrelated layer from the pinned result files. Peak selection is
 70 descriptive for this table; the full trajectories are included in the supplement and are not treated as
 71 independent replications across layers.

72 Purpose decodability is therefore reproducible under held-out cue vocabulary and remains high under
 73 cross-format transfer. This is evidence for a stable signal in the protocol. It is not evidence that the
 74 signal is a uniquely semantic representation of evaluation status. The confounded regime illustrates

Table 1: Pinned full320 observational results at the decorrelated AUC peak.

Model	Layer	Purpose AUC	Purpose BA	B→C AUC	C→B AUC
Llama-3.1-8B	8	.9888	.9409	.9513	.9401
Llama-3.1-70B	18	.9984	.9788	.9846	.9850
Llama-3.3-70B	18	.9959	.9659	.9845	.9787

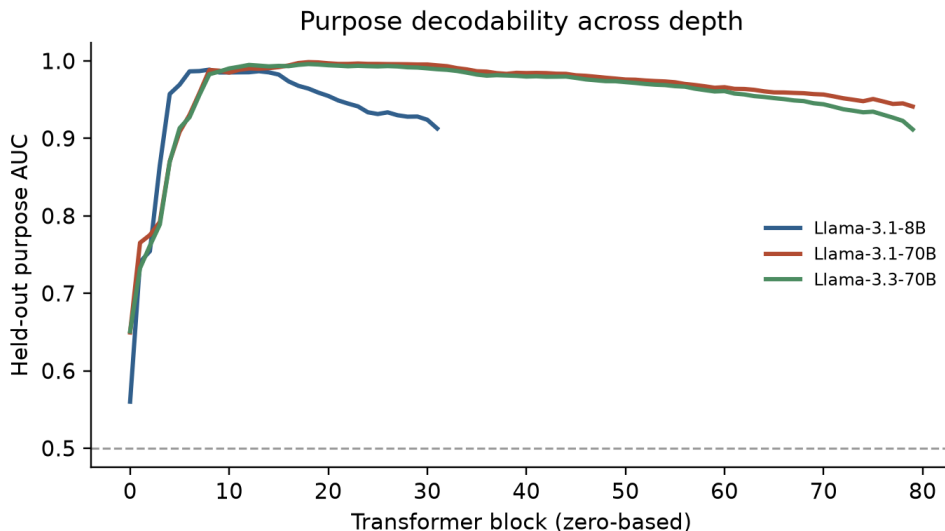


Figure 1: Held-out purpose AUC across all layers using the pinned full320 result arrays. The figure is OBSERVATIONAL and does not establish a causal or awareness construct.

75 why endpoint choice matters: its AUC can exceed .60 while balanced accuracy remains close to
76 chance at the model peaks. A score that ranks examples well but fails at its operating threshold is not
77 interchangeable with a reliable decision rule.

78 4 Protocol audits

79 4.1 Surface and format controls

80 The cue-excision surface battery removes the intended purpose words and reports near-chance purpose
81 AUC for its four purpose controls. This is a useful negative control for direct lexical copying. It does
82 not remove every possible surface or task-structure feature: format, template identity, and contract
83 identity remain intentionally decodable. In the full320 surface audit, format AUC is approximately
84 .798, template-ID accuracy is approximately .908, and contract accuracy is approximately .983.
85 These values show that the protocol contains designed structure beyond the purpose label.

86 4.2 Equal-training-size control

87 The raw cross-format comparison can be confounded by training-row count. We therefore imple-
88 mented a fixed-seed equal-training-size mixed-format control at 8B layer 12. For each held-out
89 purpose-family fold, benchmark-only, casual-only, and mixed-format regimes use the same number
90 of training payload blocks; the mixed regime balances format assignments over blocks. Complete
91 payload blocks are the bootstrap unit.

92 This control removes one important alternative explanation for the 8B result: an unequal number of
93 training rows alone is not required for high held-out purpose decodability. It does not validate the

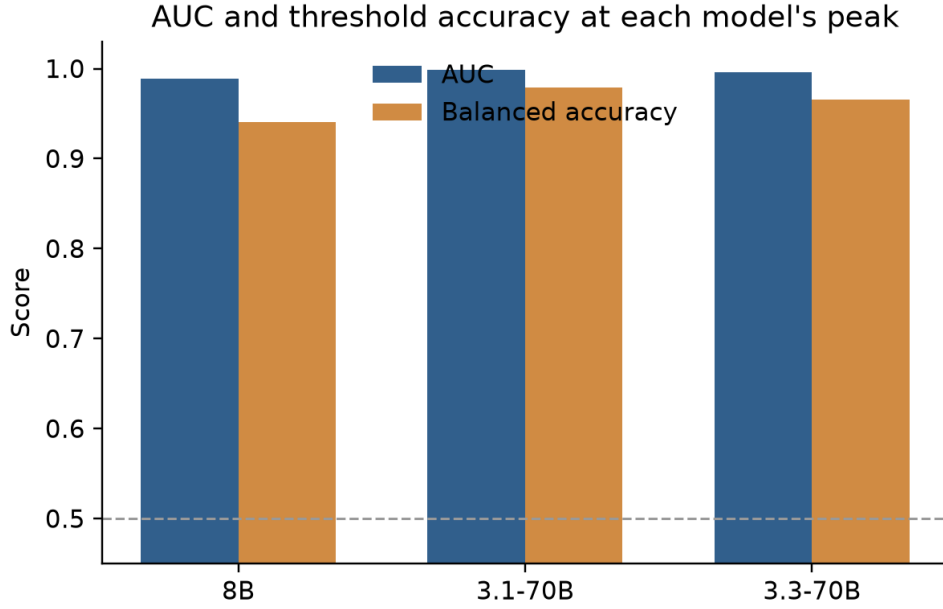


Figure 2: AUC and balanced accuracy at each model’s descriptive peak. Keeping these endpoints separate exposes calibration differences.

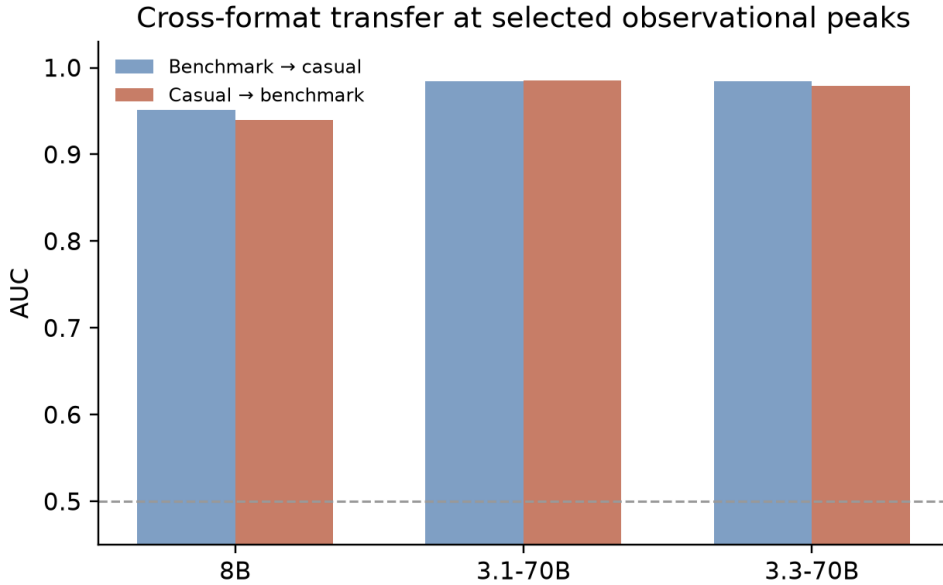


Figure 3: Cross-format transfer at each model’s descriptive observational peak. Transfer is a protocol property, not evidence of a causal awareness variable.

94 construct, and it has not yet been run for the two 70B models. Accordingly, we make no all-model
 95 equal-N claim.

96 5 Direct mechanistic diagnostic

97 We fit a model-specific Jacobian lens on 1,000 generic FineWeb prompts for Llama-3.1-8B and apply
 98 it to the full320 rows. The lens is FP32, uses source layers 4, 9, 12, 15, 20, and 24, and targets

Table 2: 8B equal-training-size control at layer 12. Values are means over four fixed seeds; confidence limits are means of the 1,000 block-bootstrap limits.

Training regime	AUC	BA	AUC 2.5%	AUC 97.5%
Benchmark-only	.9588	.8949	.9553	.9625
Casual-only	.9538	.8775	.9502	.9575
Equal-N mixed	.9797	.9241	.9773	.9821

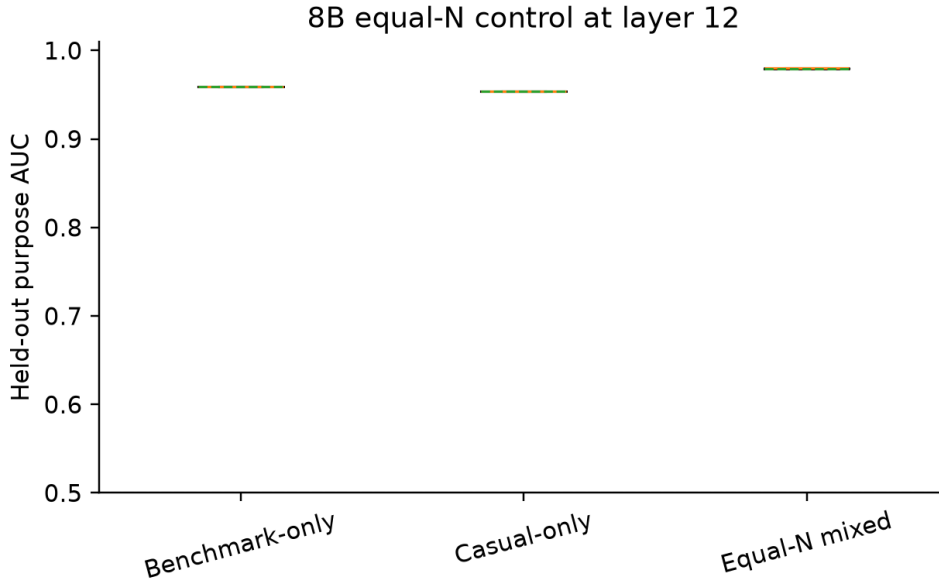


Figure 4: 8B equal-training-size control. This is a development control for one model and one predeclared layer, not a confirmation result.

99 layer 31. The assay is explicitly DEVELOPMENT-ONLY: it scores scalar concept-logit differences for
100 evaluation/deployment concept groups and a no-cue format diagnostic.

101 The oriented purpose AUC across source layers 4, 9, 12, 15, 20, and 24 is approximately .513, .569,
102 .612, .577, .608, and .576. The corresponding no-cue format AUC reaches approximately .622, .570,
103 .622, .671, .534, and .559. Thus the direct purpose readout is much weaker than the conventional
104 held-out probe, while format remains visible. A scalar factorial bootstrap over 320 blocks gives
105 positive A at several later source layers but small C relative to A ; $C/|A|$ intervals are unstable and
106 often overlap zero.

107 This result does not falsify all internal mediation hypotheses. It does show that the conventional probe
108 result and the direct lens result are not interchangeable. We therefore report the lens as a mechanistic
109 diagnostic, not as evidence of spontaneous evaluation awareness or behavioral steering.

110 6 What the protocol supports

111 The evidence supports four bounded conclusions. First, stated response purpose is decodable from the
112 frozen model representations under held-out cue-vocabulary evaluation. Second, the signal transfers
113 across the two response formats in the conventional probe. Third, format and template structure are
114 independently visible and threshold calibration matters. Fourth, a direct 8B lens diagnostic does not
115 reproduce the strength of the conventional purpose probe.

116 The evidence does not support a claim that models possess a latent evaluation-awareness variable, that
117 a purpose direction causally changes behavior, or that the effect scales to Llama-3.1-405B. No 405B

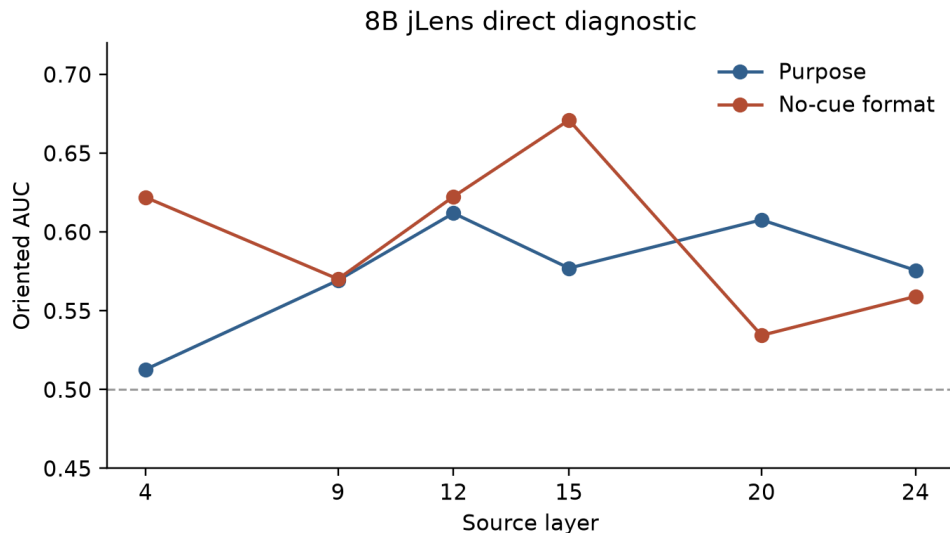


Figure 5: Direct 8B Jacobian-lens diagnostic. Purpose and no-cue format readouts are both DEVELOPMENT-ONLY; neither is a behavioral endpoint.

118 activation bundle or model revision is available in the frozen evidence package. The independent
 119 devbunova 2x2 dataset has no matched local activations and is used only as provenance/context in the
 120 appendix. The pilot80-v2 data are a separate exploratory freeze and are not pooled with full320.

121 7 Limitations and reporting checklist

122 The primary limitation is construct validity: a linear probe can recover information correlated with
 123 a prompt manipulation without identifying the intended psychological or computational construct.
 124 The equal-N control is currently 8B-only. The direct lens is also 8B-only, scalar, and development-
 125 only. We do not have an independent validated behavioral endpoint whose unsteered accuracy and
 126 calibration would justify causal intervention. Layers are repeated measurements within a model, not
 127 independent samples. Finally, the local regenerated result arrays differ bitwise from the pinned HF
 128 result files; all reported full320 numbers in this paper use the pinned result copies and their recorded
 129 revision/hash chain.

130 Our reporting checklist is therefore: define the construct before inspecting outcomes; report the
 131 complete layer trajectory; separate AUC from threshold accuracy; preserve the purpose–format
 132 factorial decomposition; use payload blocks as the uncertainty unit; report equal-N status; distinguish
 133 observational probes, geometry, internal mediation, and behavioral causality; and publish item,
 134 activation, model, code, and output hashes.

135 8 Conclusion

136 An evaluation protocol can be highly reproducible and still be insufficient for the claim attached to
 137 it. The full320 audit demonstrates strong stated-purpose decodability, but also exposes format
 138 structure, calibration failures, and a weaker direct mechanistic diagnostic. The practical lesson is not
 139 that evaluation-context probes are useless. It is that their outputs must be treated as measurements
 140 requiring construct, sampling, calibration, and causal validation—not as awareness detectors by
 141 default.

142 **A Reproducibility inventory**

143 The canonical dataset is pinned at format-EA/full1320-v3 revision
144 d6c3be7be83c469d9a6a2aa4e797f3a7d... and contains 11,520 rows from 320 payload blocks.
145 Full activation and result hashes are recorded in provenance/full1320-v3/hf_snapshot.json.
146 The submission evidence inventory is artifacts/submission.sprint.manifest.json; the
147 figure inventory is artifacts/submission.figure.manifest.json. The equal-N output
148 is artifacts/full1320_equal_n_llama31_8b_l12.json. All current outputs are marked
149 DEVELOPMENT-ONLY.

150 **B Additional pilot and dataset context**

151 The pilot80-v2 activation freeze contains 1,760 rows and is anchored by item hash
152 0dbcb9802924d3c2063fce87e24250d1a3430fd24856ea83b132e606843f6638. It is reported
153 separately as exploratory context. The devbunova parquet files are independent data with different
154 identifiers and schemas; because no matched activations are present, they are not used as a robustness
155 replication.